# SOFTWARE REQUIREMENTS SPECIFICATION

## for

## <Project>

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**<Organization>**

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# Revision History

| Name | Date | Reason For Changes | Version |
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| J.-M. Bruel | 2021-01-22 | First Draft | 1.0 |
| J.-M. Bruel | 2023-01-28 | Check after publication of the Handbook | 1.23 |
| J.-M. Bruel | 2023-06-12 | Add reqs automated numbering | 1.23.1 |
| J.-M. Bruel | 2023-08-25 | Add Minimum Requirements Outcome Principle | 1.23.8 |
| J.-M. Bruel | 2023-12-22 | Remove section numbers | 1.23.12 |
| J.-M. Bruel | 2024-08-01 | Add warning about non empty chapters | 1.24.8 |

This document follows the requirements documentation structure presented in the Handbook of requirements and business analysis, by Bertrand Meyer.

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# Goals

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**Comment:** *Goals are "needs of the target organization, which the system will address". While the development team is the principal user of the other books, the Goals book addresses a wider audience: essentially, all stakeholders.*

## G.1 Context and overall objective

**Comment:** *High-level view of the project: organizational context and reason for building a system.*

The McMaster Troitsky club is a student-run organization that registers and supports 4-6 teams of students as they take on the task of modelling, ana- lyzing and manufacturing a bridge out of popsicle sticks, white glue and dental floss. While the club has been successful in the past, results often vary in perfor- mance both between years and teams, due to the lack of centralized information and technical support for members. As teams progress through several years of competition, undergoing material testing, bridge design and analysis, and even experimenting with more innovative concepts, skill and knowledge within the club often consolidates within more experienced members or single teams, ensuring that when those members graduate, information is lost to the club. This project seeks to create a centralized information repository and technical support system for the McMaster Troitsky club, ensuring information trans- fer between years and teams, and facilitating support and onboarding for new members.

## G.2 Current situation

**Comment:** *Current state of processes to be addressed by the project and the resulting system.*

**Requirement 1.** *This is a requirement example. It refines* **??**

## G.3 Expected benefits

**Comment:** *New processes, or improvement to existing processes, made possible by the project's results.* **Comment:** *This chapter should not be empty!*

## G.4 Functionality overview

**Comment:** *Overview of the functions (behavior) of the system. Principal properties only (details are in the System book).*

## G.5 High-level usage scenarios

**Comment:** *Fundamental usage paths through the system.*

## G.6 Limitations and exclusions

**Comment:** *Aspects that the system need not address.*

## G.7 Stakeholders and requirements sources

**Comment:** *Groups of people who can affect the project or be affected by it, and other places to consider for information about the project and system.* **Comment:** *This chapter should not be empty!*

# Environment

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**Comment:** *The Environment book describes the application domain and external context, physical or virtual (or a mix), in which the system will operate.*

## E.1 Glossary

**Comment:** *Clear and precise definitions of all the vocabulary specific to the application domain, including technical terms, words from ordinary language used in a special meaning, and acronyms. This chapter should not be empty!*

## E.2 Components

**Comment:** *List of elements of the environment that may affect or be affected by the system and project. Includes other systems to which the system must be interfaced.*

## E.3 Constraints

**Comment:** *Obligations and limits imposed on the project and system by the environment.* **Comment:** *This chapter should not be empty!*

## E.4 Assumptions

**Comment:** *Properties of the environment that may be assumed, with the goal of facilitating the project and simplifying the system.*

## E.5 Effects

**Comment:** *Elements and properties of the environment that the system will affect.*

- Centralization of development information - the wealth of development information (e.g. past designs, material testing information) will be in one central location for future members to access, which will facilitate onboarding.
- Model analysis - team members will be able to submit their models and recieve key metrics and predictions regarding that model through an automatic analysis.
- Facilitated comparison - team members will be able to easily compare modelling results for different models. This will facilitate development because team members will be able to start from the best and abandon changes that caused the model to degrade.
- Simplified knowledge transfer - veteran team members will be able to share their vast knowledge with new members more easily. This is critical, especially for outgoing team members.
- Creation of a permanant software sub-team - this project will be the genesis of a permanant sofware subteam of the club. Furthermore, brief but top-tier resources alongside the software suitewill be developed to smooth the onboarding of new software subteam members.

## E.6 Invariants

**Comment:** *Properties of the environment that the system's operation must preserve.*

- The knowledge entered into the system, including development knowledge, scientific and engineering methods, and veteran wisdom, shall be fully preserved.
- Any model entered into the system, whether developed before or after the system's released, shall be fully preserved and not corrected at all to preserve development history as faithfully as possible for easy comparison.
- The rules of the competition that are relevant to the system shall be fully preserved and not be mutated at all. This is because modelling will be based of the rules as they are, and any deviation from the rules will result in inferior models.

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**Comment:** *The System book refines the Goal one by focusing on more detailed requirements about the system under development, mainly its constituents, behaviors and properties.*

## S.1 Components

**Comment:** *Overall structure expressed by the list of major software and, if applicable, hardware parts.*
**Comment:** *This chapter should not be empty!*

## S.2 Functionality

**Comment:** *One section, S.2.n, for each of the components identified in S.2, describing the corresponding behaviors (functional and non-functional properties).* **Comment:** *This chapter should not be empty!*

## S.3 Interfaces

**Comment:** *How the system makes the functionality of S.2 available to the rest of the world, particularly user interfaces and program interfaces (APIs).*

## S.4 Detailed usage scenarios

**Comment:** *Examples of interaction between the environment (or human users) and the system: use cases, user stories.*

## S.5 Prioritization

**Comment:** *Classification of the behaviors, interfaces and scenarios (S.2, S.3 and S.4) by their degree of criticality.*

## S.6 Verification and acceptance criteria

**Comment:** *Specification of the conditions under which an implementation will be deemed satisfactory.*

# Project

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**Comment:** *The Project book describes all the constraints and expectations not about the system itself, but about how to develop and produce it.*

## P.1 Roles and personnel

**Comment:** *Main responsibilities in the project; required project staff and their needed qualifications.*

## P.2 Imposed technical choices

**Comment:** *Any a priori choices binding the project to specific tools, hardware, languages or other technical parameters.*

## P.3 Schedule and milestones

**Comment:** *List of tasks to be carried out and their scheduling.* **Comment:** *This chapter should not be empty!*

## P.4 Tasks and deliverables

**Comment:** *Details of individual tasks listed under P.3 and their expected outcomes.* **Comment:** *This chapter should not be empty!*

## P.5 Required technology elements

**Comment:** *External systems, hardware and software, expected to be necessary for building the system.*

## P.6 Risks and mitigation analysis

**Comment:** *Potential obstacles to meeting the schedule of P.4, and measures for adapting the plan if they do arise.*

- **Risk: 3D Model Import and Analysis is Infeasible**
  - **Description:** Parsing STEP (.stp) files, rendering them, and extracting member geometry for the rule compliance and performance prediction tools is the most technically complex feature of the Toolbox. Available libraries may not support the required data extraction, or may not run in a web environment.
  - **Mitigation:** Prove the import and visualization pipeline in the Proof of Concept demonstration before dependent tools are built. Use an established geometry library rather than a custom parser, and agree on a standard CAD export template with the club to limit the variety of input files.
  - **Contingency:** Fall back to manual or spreadsheet entry of member geometry so the rule compliance and prediction tools remain usable without 3D import.
- **Risk: Insufficient Legacy Club Data**
  - **Description:** Past designs, material testing results and documentation are scattered across members and inconsistent in format, so the information repository may launch with little useful content.
  - **Mitigation:** Collect existing documents with the club executives early in the project and define standard upload templates for designs and material tests.
  - **Contingency:** Launch with the current season's data only, and plan for the repository to be populated over time through the upload tools.
- **Risk: Technical Issues with External Libraries**
  - **Description:** Technical issues may arise with external technologies used by tools such as bridge modelling and rule compliance, including licensing or deployment limits (for example, Autodesk APIs requiring a desktop installation).
  - **Mitigation:** Confirm licensing and deployment constraints of each library before adopting it. Team members test and validate their code before opening a pull request, and at least one other member reviews it.
  - **Contingency:** Replace the library with an open-source alternative, or reduce the affected tool's scope.

- **Risk: Unrealistic Schedule or Workload**
  - **Description:** Tasks may take longer than expected or workload may become unbalanced among team members.
  - **Mitigation:** Track tasks and milestones on the GitHub Kanban board. The Notetaker records timelines at check-in meetings, and the team revises estimates and reassigns tasks as needed.
  - **Contingency:** Because tools are largely independent, defer lower-priority tools to the software subteam's backlog rather than delaying core features.
- **Risk: User Interface is Not User-Friendly**
  - **Description:** The interface may not be intuitive for Troitsky members with varying levels of technical and modelling experience.
  - **Mitigation:** Run usability testing sessions with both new and experienced members (see the Usability Testing CD), gather feedback, and make iterative improvements.
  - **Contingency:** Prioritize fixes to the most-used workflows and add in-app guidance for complex tools.

## P.7 Requirements process and report

**Comment:** *Initially, description of what the requirements process will be; later, report on its steps.*